

Foundations of Data Analytics.

1. Database:

It is an organised collection of data that is stored in a structured way so it can easily be accessed, managed and updated

why there are essential?

because they enable business to store, manage and use massive amounts of data efficiently & securely

example: customer orders and payments

2. Data warehouse

It is a system used to store, organise, clean and structure **LARGE** data that is ready for analysis

difference between database and data warehouse?

A database is used to run the/a business (every day activities) and data warehouse is used to analyse data and make decisions

database

- used for day to day operations
- use OLTP (Online Transactional Processing)
- CRUD - Create, read, understand, transfer

data warehouse

- used for analysis and reporting.
- uses OLAP (online analytical Processing)
- ETL - Extract, Transform, Load

Why do companies use data warehouse

To analyse large amounts of data and make better business decisions.

Main reasons to use data warehouse

- Better decision making
- Stores historical data to track performance and forecast the future
- Improves reporting

3 Data Lake

Is a system where all types of data (unstructured, semi-structured and structured) is stored in the raw format until I need to use the data

difference between data lake vs data warehouse

data lake	data warehouse
Stores raw, unprocessed data	Stores processed structured data
Accepts all types of data	Mainly structured data
Stored as is	Cleaned and organised
Flexible but messy	Highly organised and reliable

When data lake would be chosen over data warehouse

When the company needs to store large volumes of raw, varied and fast growing data for future or advanced analysis

4 Data Mart

It is a subset of a data warehouse focused on a specific line of work, department (finance, IT, sales) to analyse and report for a particular business function

how does it relate to Data Warehouse?

Data warehouse stores all company wide data (central system)

a data mart is a subset of data warehouse
who uses data mart?

Marketing teams • Finance teams • Sales teams.

5. Structured	Unstructured	Semi Structure
data is organised in a fixed format (rows and columns) eg Student records	Data that has no fixed format or organisation eg images & videos.	data that does not fit neatly into tables but still have organisation using tools eg Email

6 Data Modeling

It is the process of designing and organising data structures in a way that shows how data is stored, connected and used in a database

why it is important?

It is important because it ensures that a system is well structured, efficient and reliable before it is built
it offers:

clear structure of data

reduces data duplication

improves data accuracy and consistency

makes system efficient

7 Data Mining

It is the process of examining large sets of data to discover patterns, trends and useful information

- uses techniques like Statistics, machine learning and databases

Real world examples

Youtube and Tik Tok use data mining to

analyse what I watch like and share
Hospitals and research systems used DM to analyse
patient records.

Supermarkets use data mining to track buying trends

8 Database Management

a DBMS is software system that helps manage and
control a database. It is used to create, store
manage and control access to database. It is
the interface between users and the database.

MySQL

→ uses SQL (Structured Query Language)

Very popular for web application

Use case:

- websites and online platforms eg E-commerce website

Microsoft SQL Server

→ developed by Microsoft and has strong security and
integration with Microsoft tools

Use case

- Large companies eg banking systems, government systems

Oracle

→ Designed for very large and complex system and
has excellent performance and reliability

Use case

- Large global enterprise eg airlines

PostgreSQL

→ Supports advanced data types and analytics and
very powerful and flexible.

9 ETL

Extract, Transform and Load. It is a process used to move data from different sources into a data warehouse where it can be analysed.

Extract - Collected from different sources

eg: excel files, websites, apps

Goal? gather raw data

Transform - clean and change into usable format

- removing duplicates - fixing errors - formatting data

Goal? make data accurate and consistent.

Load - cleaned data is stored in data warehouse ready for ANALYSIS

10 **CSV** - Simple tables, small data

JSON - flexible web/app data

XML - System to system exchange

Parquet - fast big data analysis

Avro - streaming and changing data structures

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING.

11. AI

Intelligence and ability of a computer or machine to simulate human intelligence, such as learning, reasoning, problem solving and decision making

Narrow AI (weak)

designed to do one specific task
cannot think outside its purpose

eg facial recognition

General AI

AI that can think, learn and perform any intellectual tasks like a human
eg ChatGPT.

12. Machine Learning

A branch of AI where computers learn patterns from data and improve their performance over time from examples instead of following fixed instructions

Traditional Programming

- rules are written by human
- works on fixed logic
- hard to scale for complex patterns

Machine Learning

- System learns rules from data
- adapt and improve over time
- good for complex and large datasets

13. Deep Learning

Is a type of machine learning that uses AI neural networks with many layers to learn from ~~learn~~ large amounts and make decisions

AI

The big umbrella

a system that makes machine act smart

Machine Learning

Subset of AI

Machine learn from data instead of fixed rules

Deep Learning

Subset of ML and works best with very large data
eg, Speech recognition

16 Generative AI

AI that creates new content such as text, audios, videos based on patterns it has learned from existing data

how it works

Training on large data

Learning patterns

Input (prompts)

Prediction and generation

Output creations

8 Tools that use Generative AI

ChatGPT - can answer questions, help with learning

DALL-E - AI that creates images from text prompts

Midjourney - AI creating highly detailed and artistic pictures

17 Training dataset.

Collection of example data used to teach an AI model how to learn patterns and make predictions

why it is important?

Accuracy of model - for correct predictions

Avoid bias - AI might learn biased behaviour

Better performance - learn faster and more reliably

Reduced errors

biased AI in need incorrect data

wrong predictions

reinforcing stereotypes

poor performance

loss of trust

18 Computer Vision

Is how computers "look at" pictures and videos and make sense of what they contain to see and understand and interpret images/videos

how computers see and interpret
they convert an image
analyse patterns in those pixels
use AI to recognise objects
Describe the image

examples

Face recognition
Self driving cars.

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API, Tools & Developer Concepts

19 API

A set of rules that allows different software applications
to communicate with each other and exchange data.

real world analogy

Restaurants

You = customer

Kitchen = System/database

Waiter = API

why they important.

enable communication

save time

allow real time data exchange

enhance automation

20 API Key

is an unique identifier used to authenticate and control
access when a user or application connects to an API

Purpose

• Security • Access Control • Usage tracking

how API key should be stored and protected

- Store in environment variables
- Use secret management tools
- Restrict access
- Use API key restrictions
- Never expose in public code

What happens if exposed in public

unauthorised access

- data theft
- financial loss
- Service abuse
- Reputation damage

21. REST API

stand for Representational State Transfer Application Programming Interface.

Methods:

GET → used to retrieve data

POST → used to create new data

PUT → used to update existing data

PATCH → Partially update data

DELETE - remove data

22. JSON

it is lightweight data format used to store data and exchange between systems

why it is commonly used

easy to read and understand

Lightweight meaning it is fast

Language dependent

Widely supported in API

JSON example

"Student id": 101

"name": Karkho

"age": 24

"course": Data analytics

"is registered": true

23 Prompt engineering

Is the skill of designing and writing clear, specific instructions to get the best possible output from AI Tools

why it is important

- improves accuracy
- saves time
- controls output quality
- essential for productivity

eg of prompts

poor prompt: explain databases

improved prompt: explain databases for beginner and include real world examples

poor prompt: Write about AI

improved: write a 150 word explanation of AI, including its definition types and real world applications

24 MAP function

Is used in programming to apply a specific function to every item in a list and return a new set of results

what it does in data processing

transforms data quickly makes coding shorter

25. Cloud computing.

Is a delivery of computing services over the internet instead of using local machines

IaaS

Provides basic computing resources like storage & servers
eg Amazon web services

PaaS

Provides a platform for developers to build, test, and
deploy apps
eg Google app engine

SaaS

ready to use software delivered over the internet
eg Google Docs

2.6 Version Control System

Is a tool that helps you track changes to files over time,
so you can save versions, compare changes and go
back to previous versions

Git?

Is a popular VCS used by developers and QA to manage
and track changes in their projects.

Why is git used?

tracks changes Collaboration Backup and recovery
experiment safely

Commit and branch

Commit is saved version (Snapshot) of work at a specific
point in time

branch is a separate version of your Project where you
can work independently.

27 Jupyter Notebook

Is an interactive tool that allows you to write and
run code, display results and add explanations all in
one document

It is most commonly used with Python

Why it is popular:

interactive coding

combining code and explanation

data visualisation

easy experimentation.

28. Python

It is a high level, general purpose programming language used for building apps, automating tasks and especially for data analysis and machine learning

Why so popular?

easy to learn and use

large community support

versatile

powerful libraries

For Python:

• Pandas

• NumPy

• Matplotlib

• Jupyter Lab

29.

Data ethics, Governance

and controls.

29. Data Privacy.

The protection of personal data and sensitive information from being accessed, used or shared

Why it is important

Protect personal information

Prevents fraud and cybercrime

Builds trust

Legal compliance

Protect user rights

GDPR

It is data protection law that regulates how personal data is collected, stored and used

Countries where GDPR applies

EU countries

EEA countries.

3.0 Data ethics.

Moral principles and guidelines that govern how data is collected, used and stored.

Examples

Using personal data without consent, problematic because it violates privacy, and no permission was given.

Biased algorithm decisions, problematic because it creates discrimination and unfair outcomes.

Data breach or poor security, problematic because it exposes sensitive information leading to identity theft and financial losses.

3.1 Data Governance

It is a set of rules, processes, roles and standards that ensure data in an organisation is accurate, secure, consistent and properly used throughout its lifecycle.

Key components.

Data owner: accountable for data quality and usage.

Data steward: manages day to day data quality and definition.

Data custodian / IT: handles storage, systems and technical control.

3.2 Bias in AI

When an AI system produces unfair or skewed results that favor one group over another. This usually reflects pre-existing prejudice in the data used in the design or decision.

used to build model

what it means when AI is biased

when it consistently gives unfair outcomes for certain groups of people

reflects existing human prejudice or inequality in its training data

Makes decisions that are not neutral or objective

why it happens

• biased training data

• lack of diversity in data

• flawed design choices

• feedback loops

real world examples

✂️ Amazing hiring tool

- AI system to help screen job applications

The system was and became biased against women

because it learned from a male dominating hiring system.

Potential consequence of AI bias.

✂️ discrimination loss of trust in AI systems

reinforcing inequality legal and ethical issues.

financial and reputational damage.

88 data analyst

analyse data to generate insights

past data

data cleaning, excel, SQL, visualisations

data scientist

builds predictive models and finds patterns

future predictions

Stats, machine learning, python

data engineer

builds & maintains data pipeline & infrastructure

data storage

database, ETL, cloud computing.

AI engineer

Create and deploy application AI systems

Intelligent application using AI models

Machine learning, deep learning, MLOPS

34 Business intelligence

Is the processing of collecting, organising, analysing and visualising data to help businesses make better and faster decisions. It turns useful insights (dashboards, reports, charts) that helps understanding what is happening in the business.

Tools

Tableau: turns complex data in visual dashboards (charts, graphs and maps)

helps business spot patterns and trends easily

PowerBI: creates interactive dashboards and reports

connects to different data sources (Excel, cloud systems)

allows to spot trends, performance and comparison in real-time

managers quickly spots sales performance, customer trends

Personal reflection

35 the datawarehous, data lake, data mining section

because it is interesting to realise so many if not all companies use data for everything.

36 AI section. I used YouTube and chatgpt to understand

37 In this modern world, almost every industry depends on data and AI to make big decisions, improve operations performance and understand customer behaviour. because of this I believe having knowledge of data and AI will be very important for my career growth and opportunities.

References

YouTube

Chatgpt.com